

Estimating and Solving with Decimals

Estimating first, then solving decimal problems exactly

Directions

- When a problem says *estimate*, round first and then compute — do not work out the exact answer and round it at the end.
- When a problem asks for the exact answer, line up the decimal points before you add or subtract.
- Write the unit on the answer line whenever the question gives one.

1. A notebook costs \$3.89 and a pen costs \$2.15. Estimate the total cost by rounding each price to the nearest whole dollar, then adding.

Answer: _____

2. One week a runner ran 48.6 km. The next week she ran 27.3 km. Estimate her total distance for the two weeks by rounding each distance to the nearest ten kilometres, then adding.

Answer: _____ km

3. Mia had \$50 in her account. She spent \$18.75 on a book and \$6.40 on lunch. Exactly how many dollars are left in her account?

Answer: _____

4. Each bag of rice has a mass of 2.4 kg. What is the exact mass of 6 bags?

Answer: _____ kg

5. Four ribbons have lengths 0.9 m, 0.85 m, 0.905 m and 0.89 m. Write the four lengths in order from shortest to longest.

Answer: _____

6. A recipe uses 1.25 L of milk and 0.4 L of cream.

(a) Exactly how much more milk than cream does the recipe use? (b) Exactly how much liquid is that altogether?

Answer: _____ L

Answer: _____ L

7. A school buys 6 folders that cost \$4.85 each. Estimate the total cost by rounding the price of a folder to the nearest whole dollar, then multiplying.

Answer: _____

8. A ribbon 7.2 m long is cut into 4 pieces of equal length. Exactly how long is each piece?

Answer: _____ m

9. Sam worked out 4.6×3 and wrote 1.38 as his answer.

(a) Work out the exact product of 4.6 and 3. (b) Write $<$, $>$ or $=$ between your exact answer and Sam's answer of 1.38. (c) Which place value did Sam put his decimal point in by mistake?

Answer: _____

10. On a hike, Jo walked 3.75 km before lunch, 2.8 km after lunch and 4.05 km in the evening.

(a) Exactly how far did Jo walk in total? (b) Her target was 10 km. Exactly how far past the target did she go?

Answer: _____ km

Answer: _____ km

11. A tap fills a container with 0.35 L of water every second.

(a) Exactly how much water is in the container after 12 seconds? (b) That water is poured into jugs that each hold 0.6 L. Exactly how many jugs does it fill?

Answer: _____ L

Answer: _____

12. A class buys 4 packs of paper at \$6.75 each and 3 markers at \$2.40 each. The class has \$40 to spend.

- (a) Estimate the total cost by rounding each price to the nearest whole dollar. (b) Now find the exact total cost. (c) Write $<$, $>$ or $=$ between your estimate and the exact total. (d) Exactly how many dollars are left from the \$40?

Answer: _____